

5(a)	1 <u>vector</u> is female <i>Anopheles</i> (mosquito) ; 2 mosquito / (female) <i>Anopheles</i>, takes blood from infected person ; I 'bites' alone 3 (vector / mosquito / <i>Anopheles</i>) inserts / AW, saliva / anticoagulant, with, pathogen / <i>Plasmodium</i> / parasite, into (blood of) uninfected person ; R if incorrect type of pathogen	2
5(b)	B-lymphocytes / B-cells ; fuse / combine / join / merge / AW ; R hybridise / bind / bond hybridoma ;	3

Question	Answer	Marks
5(c)	<i>any three from</i> 1 results are accurate (rely on monoclonal antibodies) ; 2 results are obtained quickly / diagnosis is quick ; 3 low cost / cheap ; 4 does not need highly trained medical professional / easy to read / easy to use / AW ; 5 does not need specialised equipment ; 6 easily, transported / distributed (in bulk) ; 7 can ensure appropriate treatment is given immediately malaria is confirmed ; 8 reduces unnecessary use of antimalarial drugs (just in case) ; 9 (a small area for S so) only a small sample of blood needed ; 10 AVP ; e.g. can identify different species of <i>Plasmodium</i> I ref. to line C and working correctly	3

Question	Answer	Marks
5(d)	ignore ref. to sickle cell anaemia <i>any five from</i> <i>habitat of Anopheles mosquito</i> 1 tropical / warm and humid / AW, climate / regions ; A in context of global warming (increases life span of <i>Anopheles</i>) 2 in areas where, <i>Anopheles</i> (mosquito) / vector, occurs ; 3 <i>Anopheles</i> mosquitoes only live in humid conditions ; 4 warm temperatures for, development / growth, of, parasite / <i>Plasmodium</i> (in vector / mosquito) ; 5 warm temperature for, development / growth, of mosquito larvae ; 6 mosquitoes require bodies of, still / AW, water for breeding ; A ponds, puddles, lakes, swamps 7 mosquitoes require places where there is sufficient rainfall ; 8 low altitude ; <i>resistance</i> 9 insecticide / repellent, resistance of mosquitoes ; 10 drug resistance of parasite ; <i>prevention</i> 11 in countries / areas, where, prevention / control measures, are not implemented by, governments / health authorities ; 12 further detail e.g. any example of a, prevention / control measure that is not used or not implemented fully ; <i>immunity</i> 13 immunity to malaria in human population (limits distribution) ; <i>people</i> 14 migration of infected people from areas (with high rates of malaria) ; 15 high rates of HIV infection ;	5